

Progettazione di Applicazioni Mobili

Interactive software design

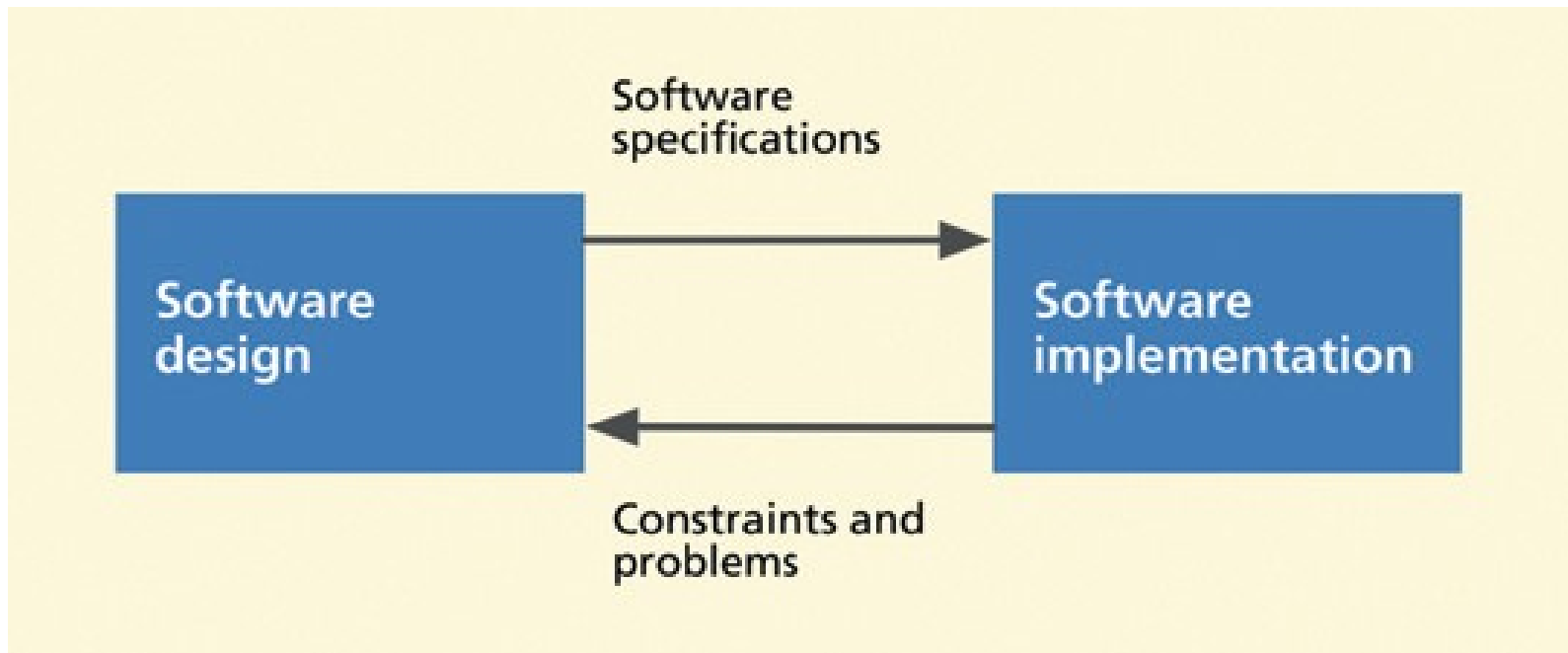
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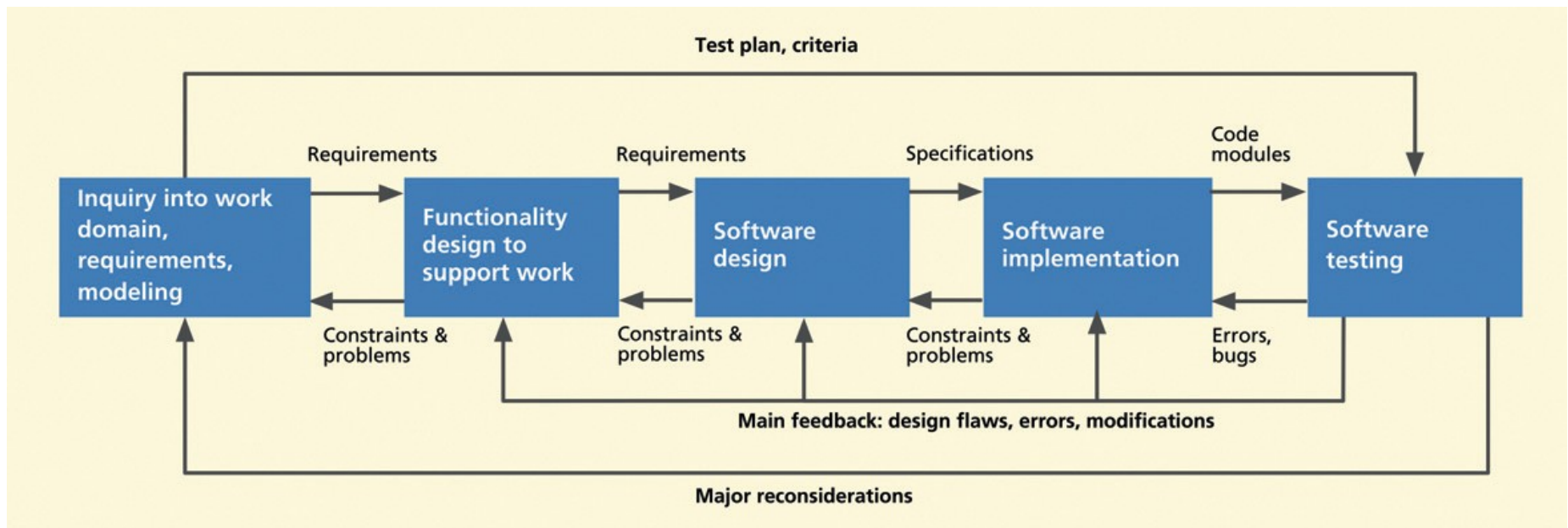
stefano.burigat@uniud.it



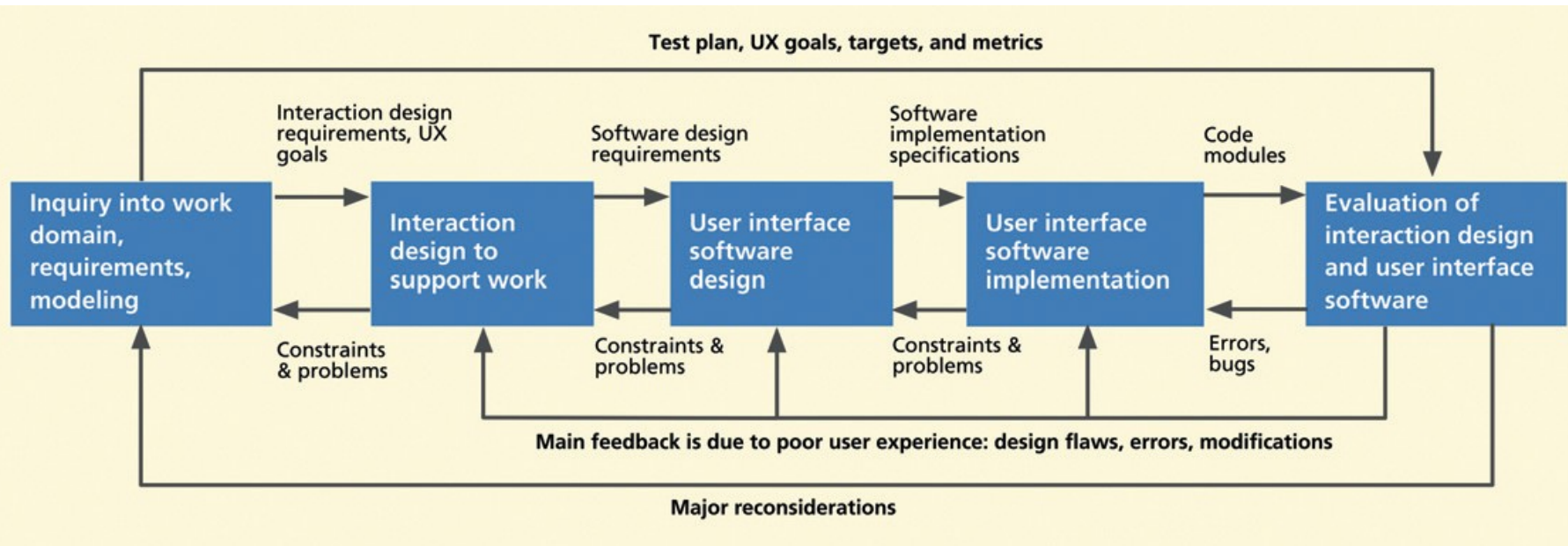
- Fundamental activities involved in building an **interactive software**

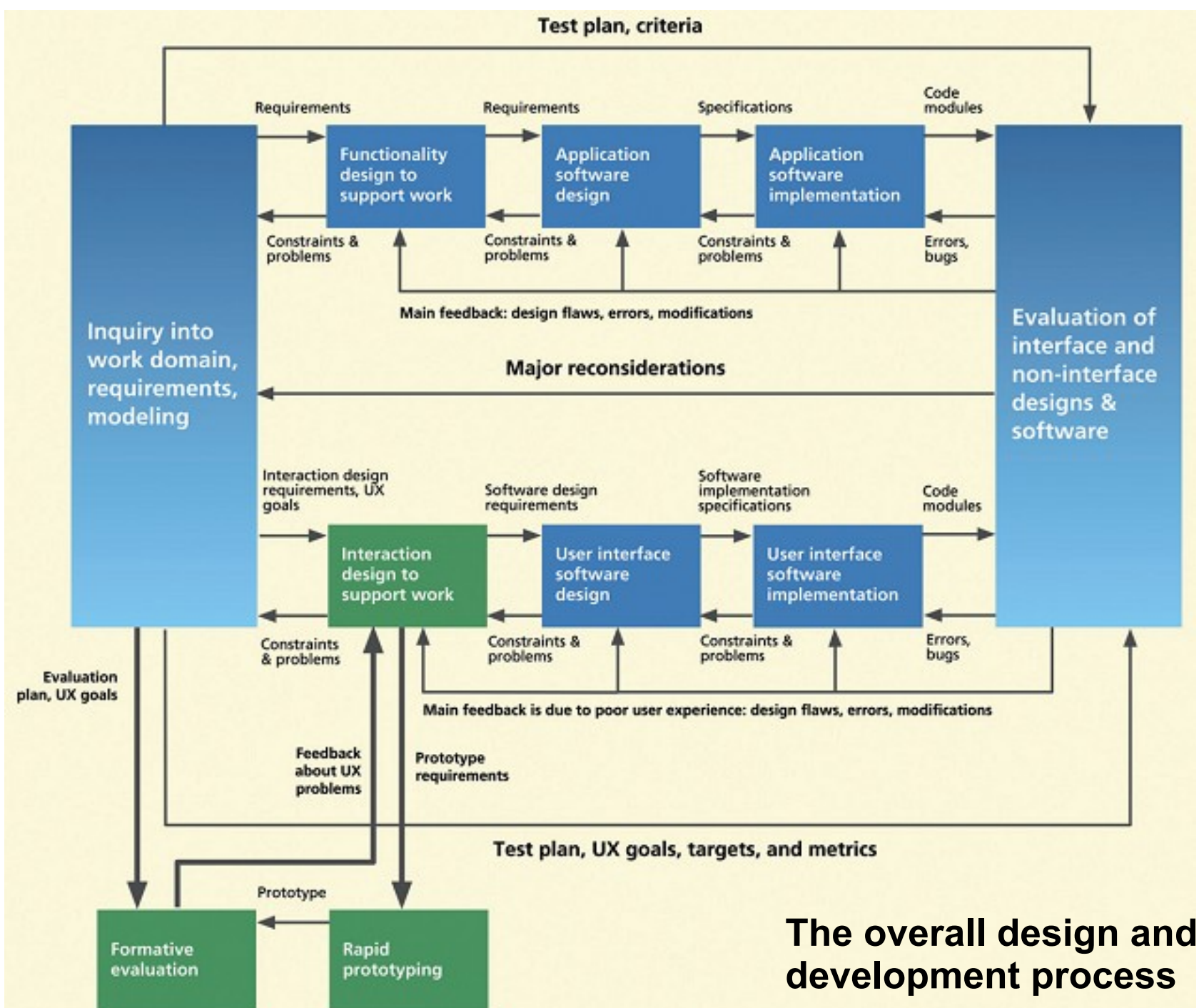


- Additional steps needed as software **complexity increases**



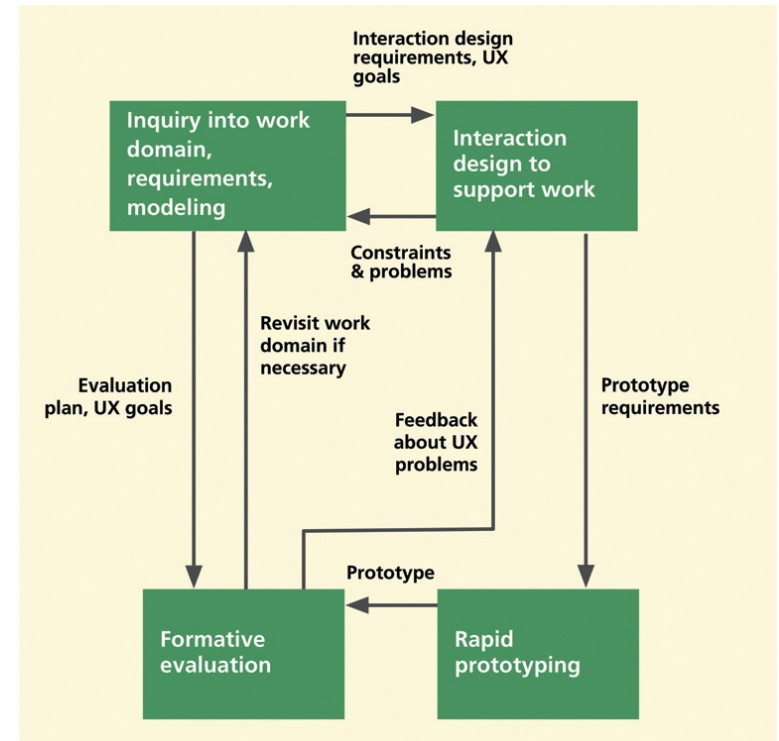
- Usually, separate but similar process for the **user interface**





The overall design and development process

- To users of an interactive software, the experience of interacting with the software is the software
- Iterating the interaction design / UX part of the whole process is a key activity to ensure a quality **user experience** in interacting with a software

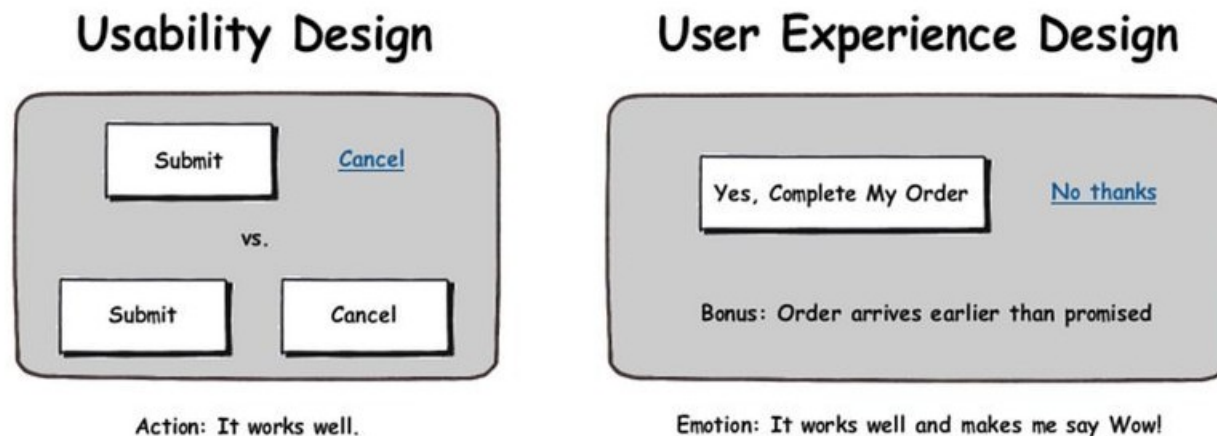


- **User experience (UX)** is the totality of the effect(s) felt by a user as a result of interaction with, and the usage context of, a software, device, or product

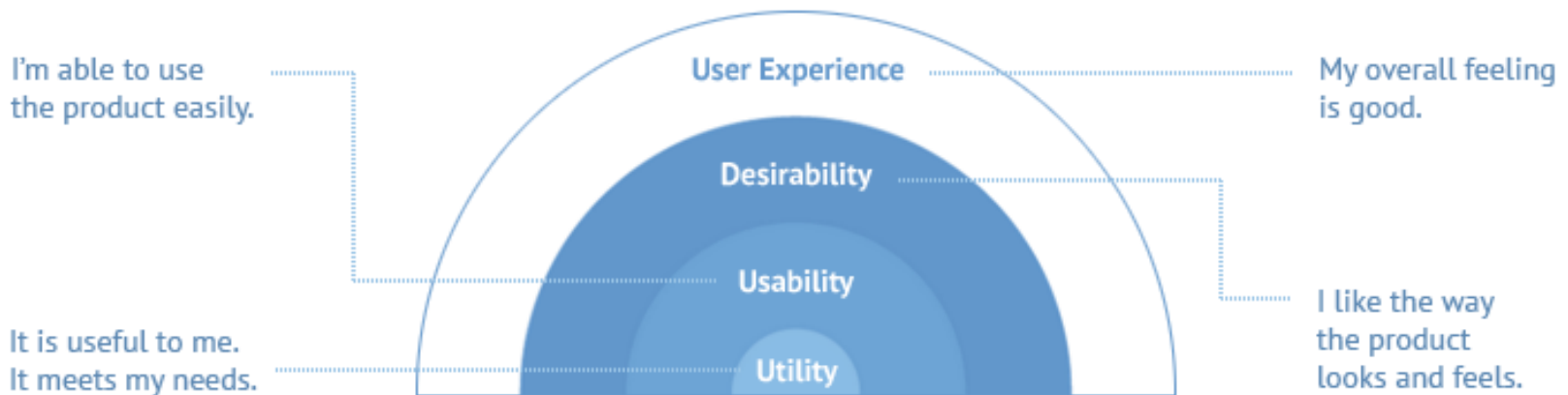


- A single design might result in multiple UXs (due to variations in users or contexts of use)
- UX is affected by usability factors, utility factors, emotional impact factors, social/cultural factors, marketing factors

- **Usability** is a pragmatic multi-dimensional concept that includes (measurable) aspects such as ease of use, productivity, efficiency, effectiveness, learnability, retainability, and user satisfaction (ISO 9241-11, 1997)



- **Utility** is about providing functions that users need in the first place
- Only when usability is combined with utility does software become really useful to users



- **Emotional impact** is the affective component of UX that influences user feelings
- Emotional impact includes such effects as pleasure, fun, joy of use, aesthetics, desirability, novelty, originality, sensations, coolness, engagement, novelty, and appeal



Visceral Emotional

- Attractiveness
- Pre-consciousness
- Initial impression
- Feelings



Behavioral Emotional

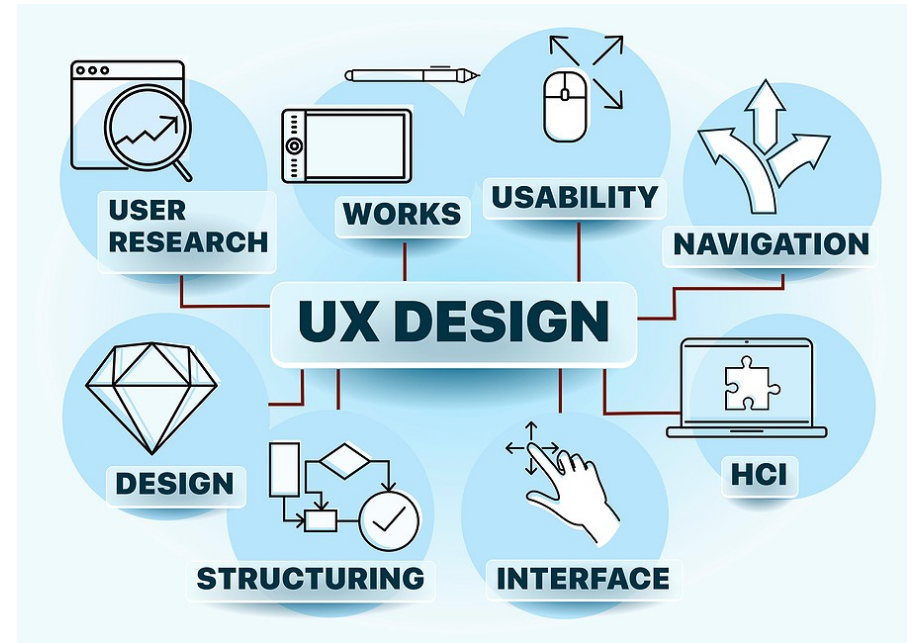
- Usability
- Product function
- Performance
- Effectiveness of use



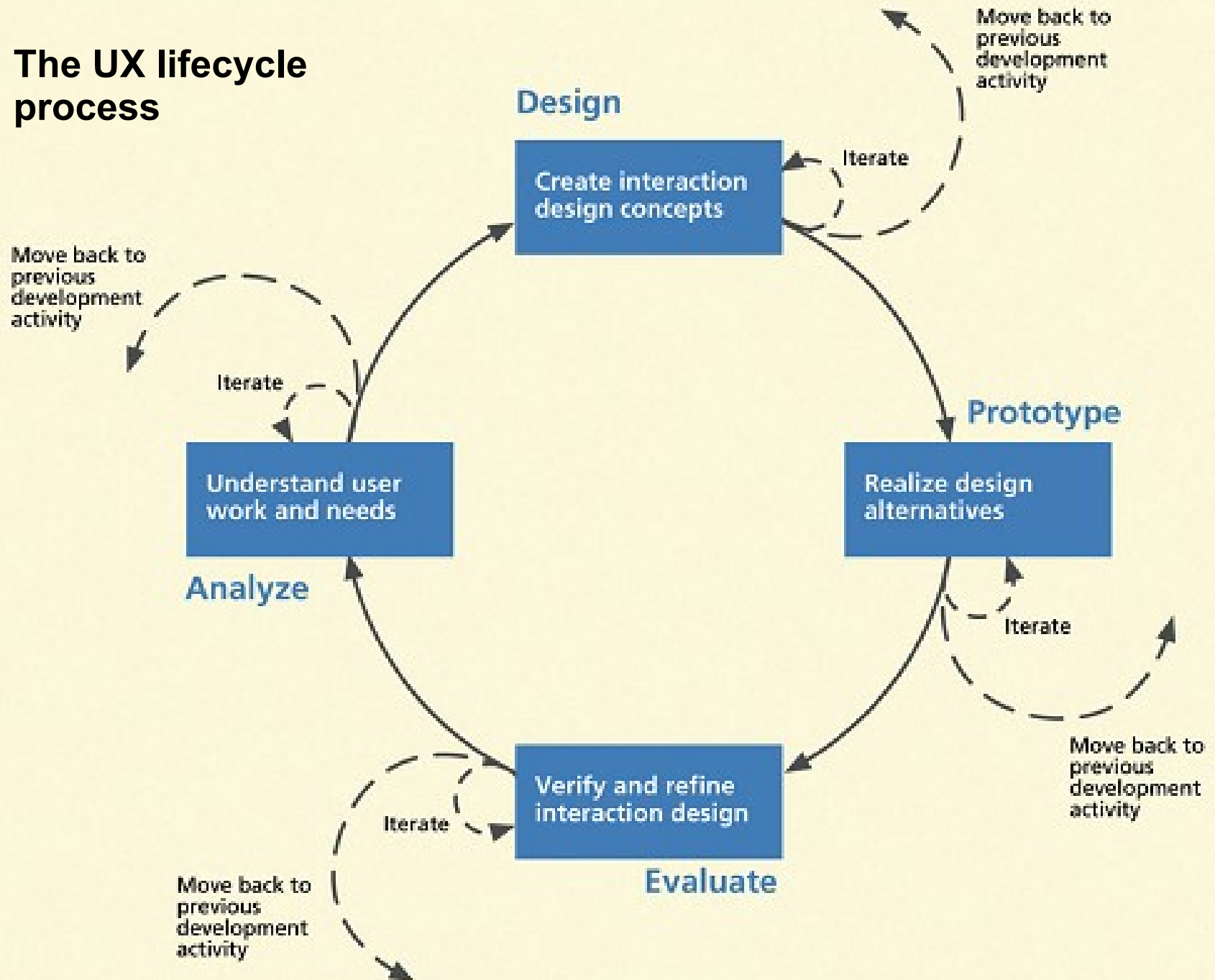
Reflective Emotional

- Meaning of product
- Impact of thought
- Sharing the experience
- Cultural meaning

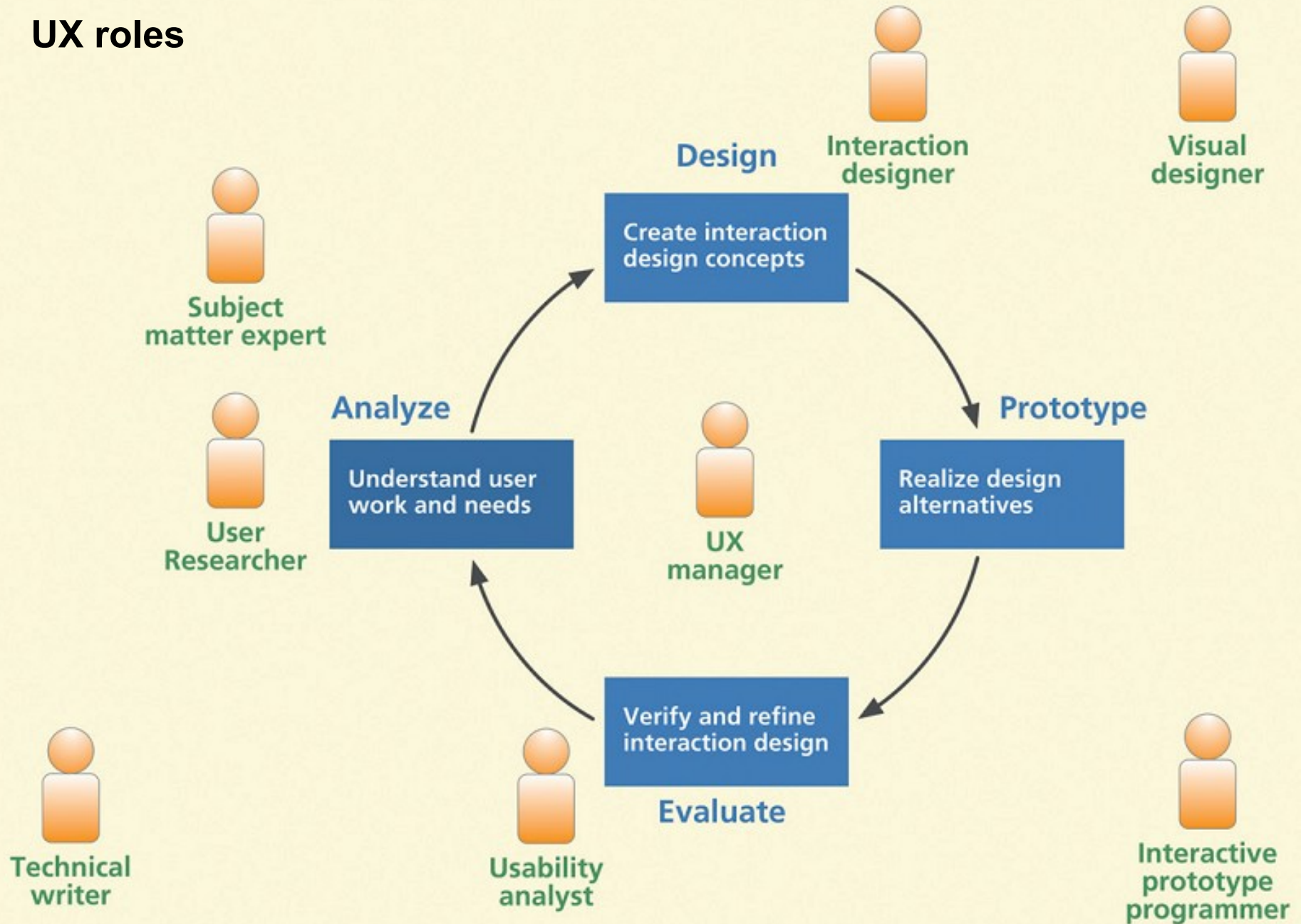
- There is **no formula** for creating an interaction design that leads to a specific user experience
- However, the more designers follow a **UX process** focused on users and usage context, the better they will be able to create a design that can lead to a desired user experience

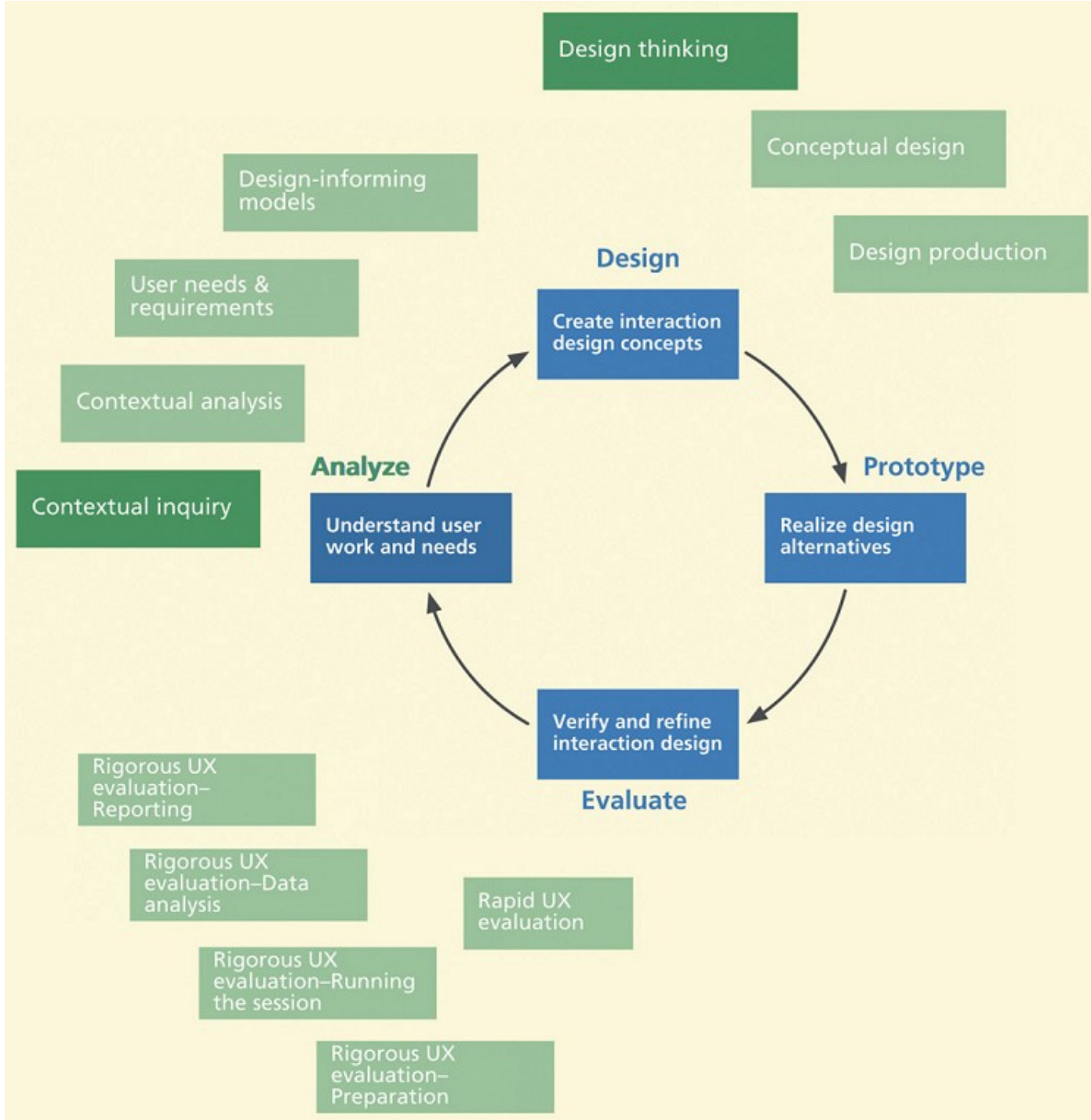


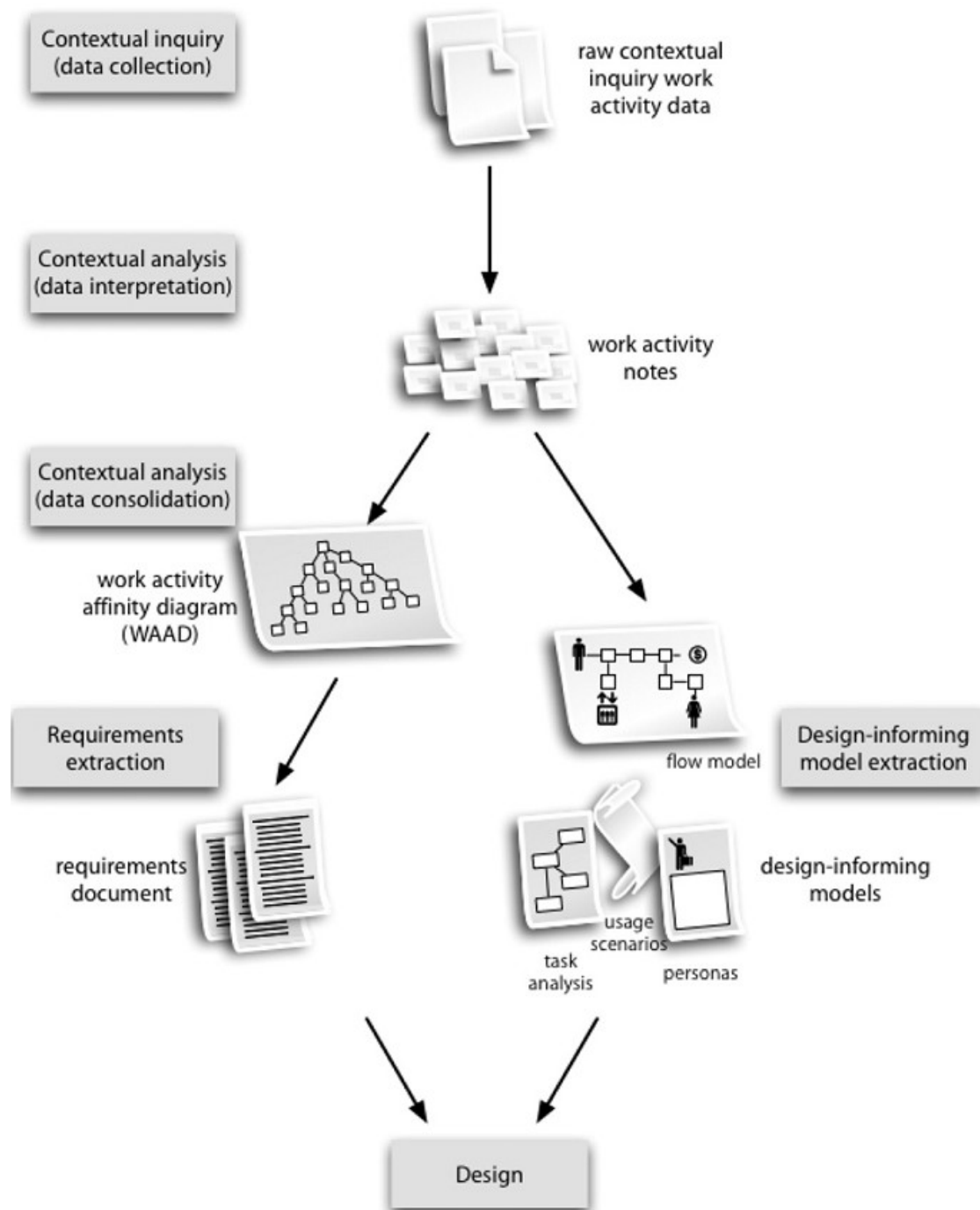
The UX lifecycle process



UX roles









CONTEXTUAL INQUIRY

UX Knowledge Base Sketch #51



CONTEXTUAL: YOU INTERACT WITH THE PARTICIPANT IN HER/HIS NORMAL ENVIRONMENT, CONTEXT.



INQUIRY: YOU ASK QUESTIONS
→ TO CLARIFY WHAT YOU'VE OBSERVED
→ TO BETTER UNDERSTAND THE TASKS



ALSO CALLED:
SITE VISIT



YOU CAN GET INSPIRATION FROM ETHNOGRAPHIC RESEARCH



ADDITIONAL METHOD:
SHADOWING



GOAL: IN-DEPTH UNDERSTANDING OF THE

- ↳ CONTEXT
- ↳ ACTIVITIES, TASKS
- ↳ PAIN POINTS, ISSUES, FRUSTRATIONS
- ↳ BEHAVIOR
- ↳ PROCESS, ROLES, INPUTS & OUTPUTS

WHILE IT IS IMPORTANT TO LEARN ABOUT PAIN POINTS, YOU SHOULD GO FOR A BALANCED SESSION: YOU ARE THERE TO UNDERSTAND HOW TASKS ARE ACCOMPLISHED (AND WHAT ISSUES OCCUR DURING COMPLETION)



KEEP THE SESSION BALANCED!



THIS UNDERSTANDING WILL HELP TO MAKE INFORMED DESIGN DECISIONS



CONDUCTING A CONTEXTUAL INQUIRY HELPS

- (IN)VALIDATING YOUR ASSUMPTIONS (E.G. IS THAT USER NEED EXIST?)
- FINDING CLUES, SIGNALS IN THE ENVIRONMENT TO POSSIBLE ISSUES (E.G. CHEAT SHEETS ON A STICKY NOTE) ⇒ POTENTIAL AREAS FOR IMPROVEMENT!
- DISCOVERING ARTIFACTS & ACTORS YOU'VE NOT THOUGHT ABOUT
- CREATING MORE REALISTIC PERSONAS, SCENARIOS, JOURNEY MAPS ETC.

DURING THE CONTEXTUAL INQUIRY



E.G. TRICKS SHORTCUTS!

BE A FLY ON THE WALL! OBSERVE, CAPTURE (EVERYTHING - AT THIS POINT YOU DON'T KNOW WHAT THINGS ARE RELEVANT!)



CONDUCT "TRADITIONAL" SHORT INTERVIEWS)

- GET AN OVERVIEW OF THE ACTIVITY
- BUILDING RAPPORT WITH THE PARTICIPANT



NAIVE QUESTIONS ARE GREAT!

- "MASTER-APPRENTICE" RELATIONSHIP
- ASK FOR DEMONSTRATION INSTEAD OF EXPLANATION
- AGREE ON THE "RULES" OF INTERRUPTION BEFOREHAND



DOCUMENTATION!
- NOTES & SKETCHES
- PHOTOS
- RECORDINGS (IF ALLOWED)



SUM UP WHAT YOU'VE OBSERVED, ASK WHETHER YOUR UNDERSTANDING IS VALID OR NOT ⇒ THE DIFFERENCE BETWEEN WHAT YOU SAW & HOW THE PARTICIPANT INTERPRETS THAT IS A VALUABLE INSIGHT!



A TOO BIG GROUP CAN DISTURB THE PARTICIPANT, HOWEVER, A TEAMMATE CAN HELP YOU CAPTURE MORE, AND YOU CAN DISCUSS YOUR IMPRESSIONS LATER.



MAIN ADVICE: GO WITH THE FLOW, YOU MIGHT GET UNEXPECTED INSIGHTS! E.G. NEW USER NEEDS



SOME CHALLENGES:

- MOTIVATING THE PARTICIPANT TO SHOW EACH RELEVANT STEPS (VS. RUSHING THROUGH)
- PRIORITIZING WHAT YOU PLAN TO OBSERVE (SO YOU DON'T RUN OUT OF TIME)
- EXPLAINING WHAT THE METHOD IS ABOUT & WHAT YOU ASK THEM TO DO
- IT CAN BE HARD FOR THE PARTICIPANTS TO SWITCH FROM A MORE PASSIVE POSITION TAKEN DURING THE INITIAL INTERVIEW TO AN ACTIVE ROLE OF PERFORMING TASKS
- MINIMIZING THE NUMBER OF OBSERVERS



PREPARATION

- SCHEDULING; PLANNING TRAVEL (TIME & COST)
- BASIC UNDERSTANDING OF THE DOMAIN (E.G. FROM STAKEHOLDERS, DOMAIN EXPERTS)
- YOUR FOCUS - IDEALLY, YOU'LL OBSERVE MANY UNEXPECTED THINGS, BUT IF YOU NEED TO OBSERVE TASKS THAT RARELY OCCUR: EXPLICITLY ASK PARTICIPANTS TO PERFORM THEM
- LISTING OUT YOUR ASSUMPTIONS
- CONSENT FORMS
- INTERVIEW SCRIPTS (E.G. FOR MULTIPLE SHORT ONES)
- TESTING / PREPARING RECORDING EQUIPMENT
- EXPLAINING THE ESSENCE OF THE METHOD: PARTICIPANTS SHOULD KNOW WHAT TO EXPECT



AFTER THE CONTEXTUAL INQUIRY: ANALYZE DATA
E.G. WITH AFFINITY DIAGRAMMING - LOOK FOR PATTERNS!



DESIGN

SEARCH



work activity
notes

Contextual analysis
(data interpretation)

Contextual analysis
(data consolidation)

work activity
affinity diagram
(WAAD)



Downside: Privacy panels would limit ability of a group to see the screen and work together on buying tickets. Conclusion: maybe not worry about it too much.

I will be concerned if the kiosk only gives me a ticket confirmation and not the ticket itself.

U4: Since we do not pay for restaurant there may be misuse maybe use credit card to prepay a penalty (in \$) for those who reservation.

Another privacy idea: Need system timeout, reset to Home page after 1-2 minutes of inactivity.

Security
Concerns
+ privacy
+ trust

U4: The kiosk should have a way to ensure the person who is calling the customer service number is actually physically located at the kiosk to prevent misuse. So the kiosk should probably display a number which the customer can read out to the customer service representative to ensure they are physically located there.

→
Printer/
Troubleshooter

U4: Identity theft and credit card fraud is a huge concern to me.

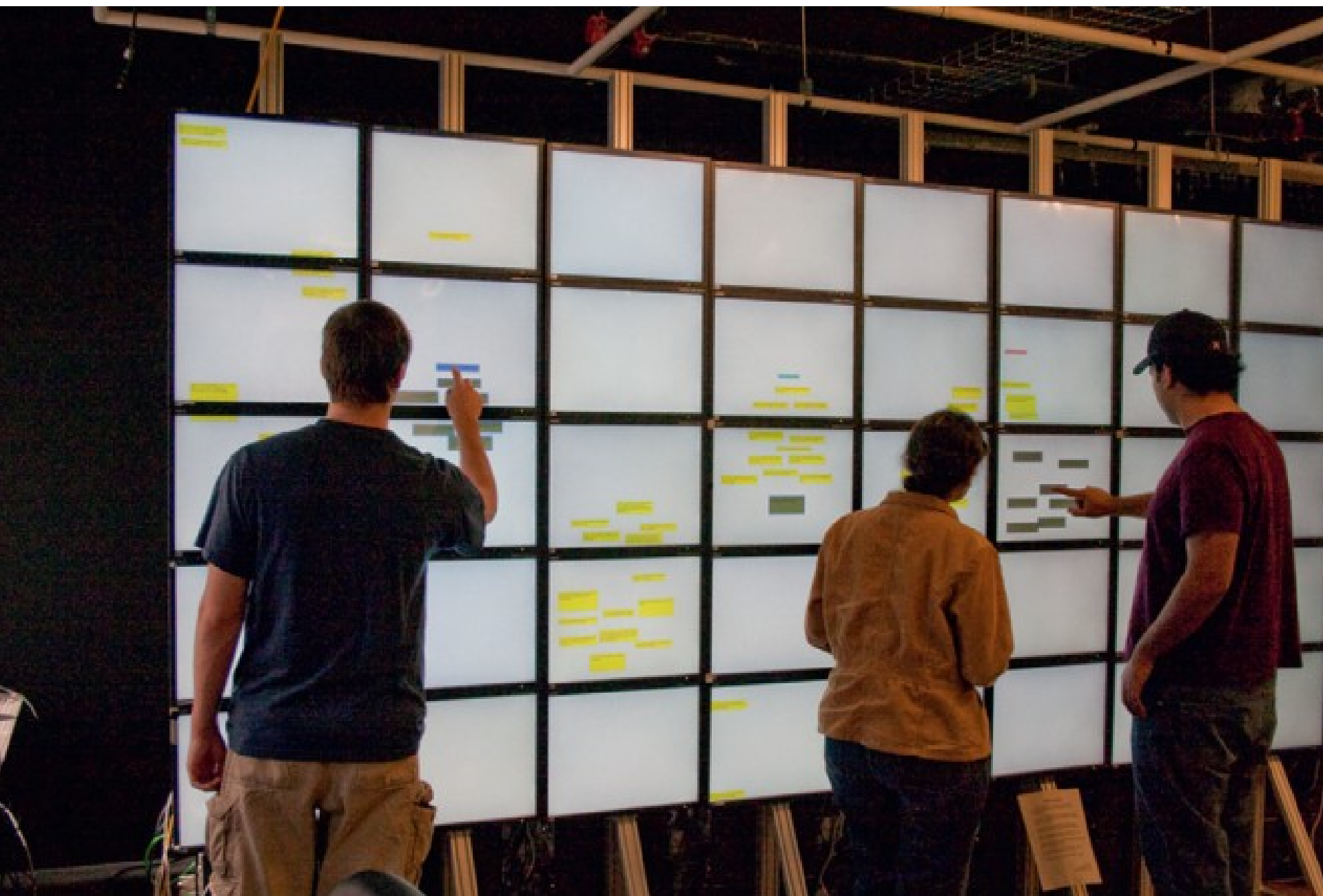
U4: I also wonder how often this expensive hardware (kiosk) will be stolen

Privacy: Sometimes I choose not to print my receipt, since I usually just throw it away (e.g., at the gas station) and don't want someone else to get it.

U4: I would trust an established company to manage my credit card transactions for restaurant reservations and such but not the kiosk itself.

U4: The kiosk should have a recognized brand name on it for me to trust it, like TicketMaster





AFFINITY DIAGRAM

GAMIFICATION

FREE WORK
SPECIAL
ONLINE
BOOKINGS
STIMULATING
THE USER'S
CREATING A
RATTLER WITH
SOME SPECIAL
ELEMENTS
POINTS?
COLLECT POINTS
FOR FURTHER
FLIGHTS

DATA ANALYSIS

WHAT IS
THE TYPICAL
PRICE FOR
THE ROUTE?
AFTER SELECTING
EXTRAS -> SMART
SUGGESTION TO
TAKE SPECIAL
CHARTER
STATISTICS
ABOUT WHICH
FLIGHTS ARE
FAVORITE

FRUSTRATION

WHY THE
SITE IS SO
JUMPY?
NO ADS
I DON'T NEED
SPECIAL
RATES!
SELECT FLIGHT
WITH NO EXTRA
OPTIONS ->
BUTTON - ADD EXTRA

ACCESSIBILITY

SCREEN READERS
AND
KEYBOARD NAVIGATION
INTEGRATION
WITH
MOBILE APP
THE SITE MUST
BE 100%
ACCESSIBLE

SIMPLICITY AND CLARITY OF THE STRUCTURE

CLEAR CTA
DATE SELECTION
PAGE
ONE CLEAR CTA PER
PAGE
CLEAR INDICATION
OF DIRECT
FLIGHTS
WHY CAN'T I
SELECT
A FLIGHT?
IS THIS THE
FINAL PRICE
OR THERE ARE
SOME ADD-ONS
CTA COPY
EXACT AS THE
WHAT THE USER
IS DOING
CAN NOT
FIND THE
SORTING
OPTION!
SEAT SELECTION
CLEAR PICTURE
OF THE SEAT
SELECTION
CLEAR
INDICATION
OF SPECIAL
RATES

USEFUL FEATURES

SUMMARY ON
THE LEFT
WHAT IS CHANGING
EXTRAS ON
ONE PAGE
WITH CHECKBOXES
ONE WEEK
SPAN FOR
DATE SELECTION
SORTING
BY PRICE, DURATION
OR FLIGHT
DURATION
CHANGES/CANCELS
BOOKINGS & DOWNS
LINK OUT TO
PILOT PAGE
FORM FOR THE
DATA OF EACH
ON ONE PAGE
EMAIL - SUMMARY
OF THE CURRENT
TRIP PAGE
OUTBOUND/
INBOUND ON
SEPARATE PAGES
EMAIL
REMINDER
FOR UPCOMING
FLIGHTS
PAYMENT INFO.
STRUCTURED
FIELD FOR THE
CREDIT CARDS
MOST IMPORTANT
IS RETURN
OPTION

LOGGED IN USER PERKS

SAVE
SEARCH
SEAT PREFERENCE
REMARKS FOR
EACH USER
IF 0 IS LOGGED
IN -> SEARCH
FIELD -> LAST
SEARCHES
MARKER
FOR
FAVORITE
DESTINATION
ADDING/LOADING
AVATARS FOR
EACH USER?
LAST BOOKINGS
IF REQUESTED
THE USER CAN
ALSO LAST SEARCH

STRUCTURAL AND FUNCTIONAL HELPFUL FEATURES

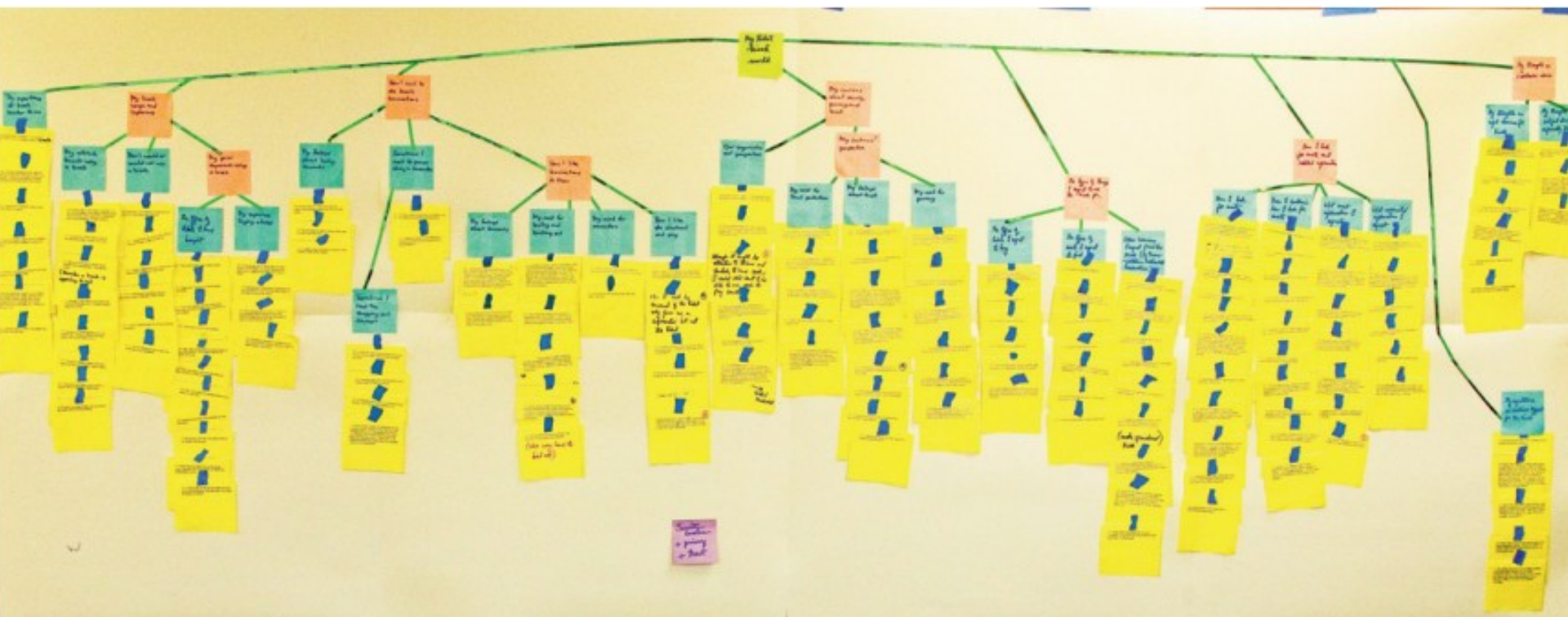
MATRIX
VIEW
OF FLIGHTS
LINK SHARING
ON CERTAIN
SEARCH/BOOKING
HOME PAGE
SPLIT THE SCREEN
IN HALF - MAP
HALF - SEARCH
FIELDS
USER CAN
SET A
BUDGET
FOR THE TRIP
I WILL PAY
MY PART OF
THE TRIP AND
SHARE/SEND
IT TO A FRIEND
INDICATION
OF CO2 FOR
FLIGHT?
I WANT TO
SEE THE
CARRIER'S FLIGHTS
FOR UPCOMING
OUTREACHES
SMART
FILTER?

STEP BY STEP APPROACH

ONE THING
PER PAGE
AFTER FOLLOWING
OF THE DATA OF
ALL PAGES -> JUST
SUMMARY -> JUST
RELEVANT PAGES
STRUCTURAL
REPRESENTATION
OF THE DATA IS
SIMPLE AND
HIGHLIGHT THAN
VISUAL
CONFIRMATION PAGE
EVERYTHING ON
ONE PAGE
TO BOOK
WITHOUT
USER
PROFILE
CREATION OF
USER PROFILE
POSSIBLE BUT
NOT A MUST

START SEARCH PAGE - NICE TO HAVE

AUTO-CORRECT
SUGGESTION
ON AUTO-COMPLETION
GEO LOCATION
ON START
SEARCH PAGE
NUMBER OF
PASSENGERS
LIKE AIR
LINES (1, 2, 3, 4, 5)
START SEARCH
MUST BE VERY
SIMPLE AND
CLEAR
BIG
PROMINENT
START
SEARCH





Project name: Online shopping system **Starting date:** 1st Jan, 2012 **Completion date:** 31st August, 2012 **Traceability matrix number #:** 1
Project description: The client wants to develop a shopping system, by which they can sell their products

Req. id	Rel. id	Functional requirements					Teams	Status	Design document	Non-functional requirements
		Main requirements	Sub requirements	Category	Description	Relationship				
1	1	Overall project	- Service - Payment	Required	There will be service and payment modules in the online system	Composition	C, RE, A (Australia) Dv (India, China)	Started (work in progress)	Use cases (1-19)	Performance, security, usability, support, availability, localizability
2	1.1	Project - Service	Purchase Order tracking Seller information	Required	Following services are required: purchase; order tracking; and seller information	Composition	Team in India	Started (work in progress)	Use case 2	Performance, security, usability, availability, localizability
3	1.1.1	Service - Purchase	Browse catalogue Select product Make payment Place order	Required	To make a purchase, the following steps are necessary: browse catalogue; select product; make payment; and place order	Composition			Use case 3	Performance, security, availability, usability
4	1.1.1.1	Purchase - Browse catalogue	Browse catalogue	Expected	To view product information	Association			Use case 4	Performance, availability
5	1.1.1.2	- Select product - Make payment	Select product	Expected	To choose required product	Association			Use case 5	Performance, availability
6	1.1.1.3	- Place order	Make payment	Expected	To pay required amount	Association, intersection			Use case 6	Performance, availability, security
7	1.1.1.4		Place order	Expected	To finalize and place order	Association			Use case 7	Performance, availability
14	1.2	Project - Payment	Payment mechanism Authentication mechanism	Required	Enable customers to make their payments	Composition	Team in China	Started (work in progress)	Use case 14	Performance, security, usability, availability, localizability
15	1.2.1	Payment -Payment mechanism	Payment via credit card	Required	Customers can pay by credit card	Association			Use case 15	Performance, security, availability, usability
19	1.2.2	Payment -Authentication mechanism	Verify card details	Required	Validation criteria associated with payment method	Association, intersection			Use case 19	Performance, security, availability

Mobile Web Requirements

Created by Mitch Davis, last modified just a moment ago

Target release	1.0
Epic	 MDT-18 - Mobile optimized web app TO DO
Document status	DRAFT
Document owner	@Mitch Davis
Designer	@Cassie Owens
Developers	@Harvey Jennings
QA	@Kevin Campbell

Background and strategic fit

We all know mobile is on the rise. A [recent survey](#) to customers showed that 85% of users use their mobile on a daily basis. Most of our customers also use competitor apps, so this is something we need to have.

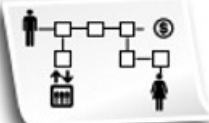
Customer research

- [Customer interview - Netflix](#)
- [Customer interview - Homeaway](#)
- [Customer interview - Bitbucket](#)

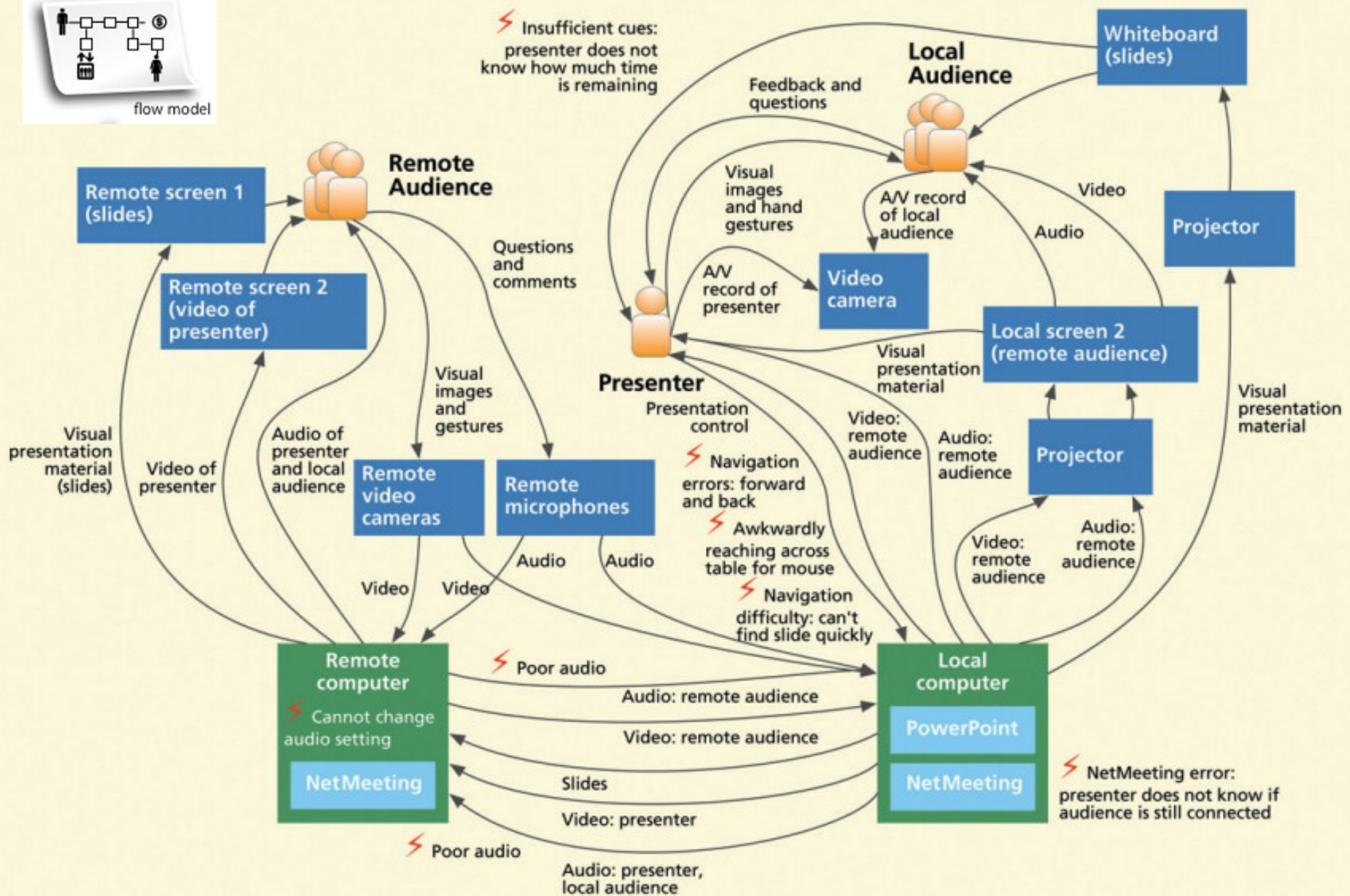
Requirements

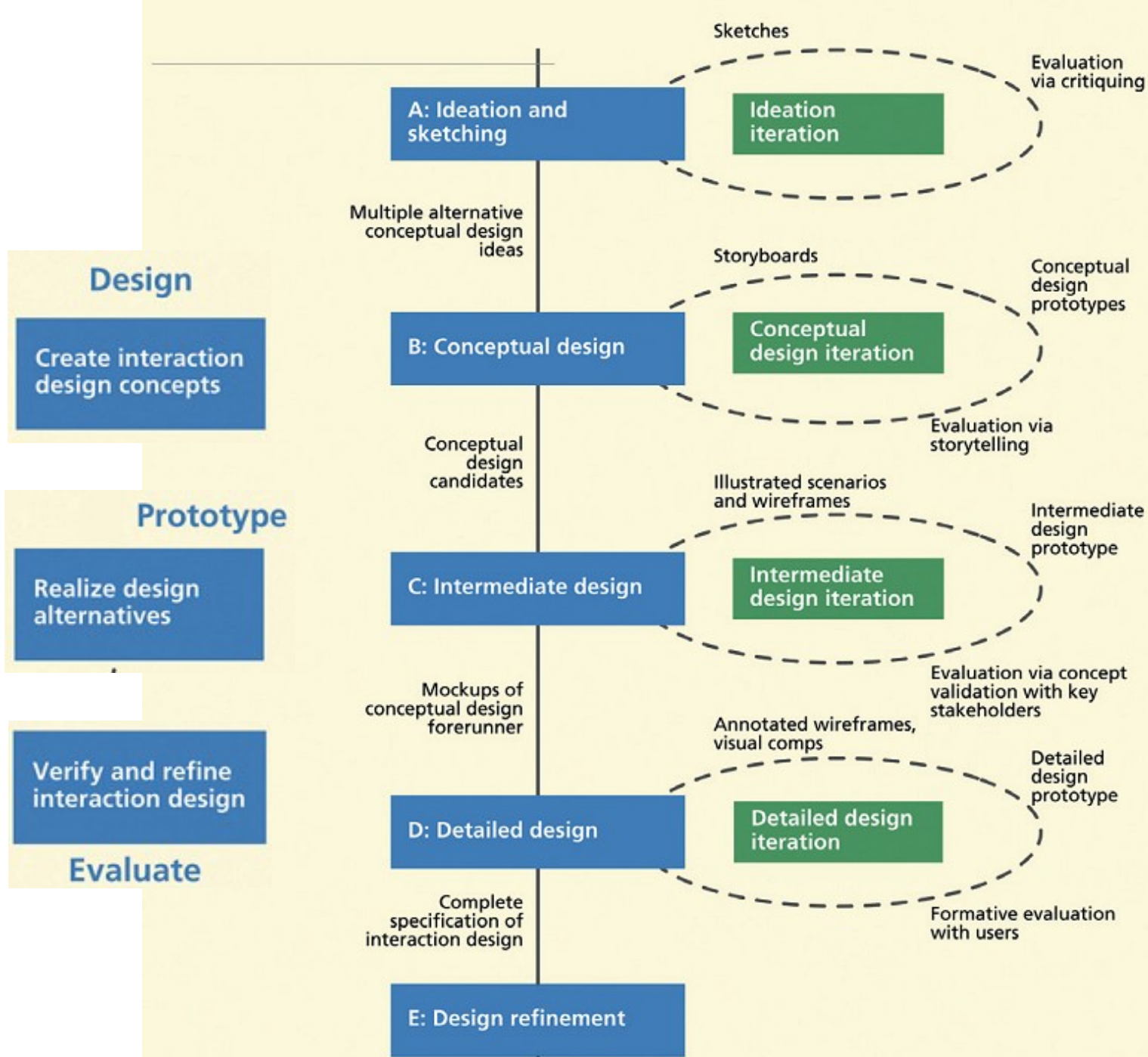
#	User story title	User story description	Priority	Notes
1	Facebook Integration  MDT-13 TO DO	A user wants to sign up via Facebook	Must Have	• We will need to talk to Cassie Owens. • There has also been some research done on this (see Facebook integration prototype)
2	Activity Stream  MDT-14 TO DO	A user wants to view the latest updates via the mobile dashboard so that they can get a better understanding of what is in place	Must Have	
3	Post Updates  MDT-15 TO DO	A user wants to be able to post status updates on the go	Must Have	The key things we will need to support: • Text status updates • Mentions • Support for images • Smart embedding for YouTube vids
4	API  MDT-16 TO DO	A developer wants to integrate with the mobile app so that they can embed the activity stream on their website	Should Have	• We should chat to Team Dyno as they did something similar.

Design-informing model extraction



flow model

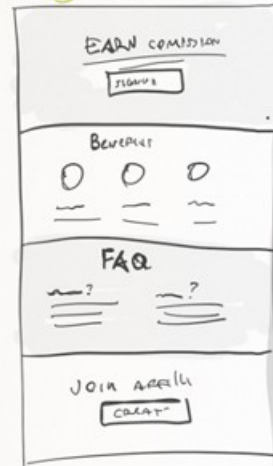






Affiliate PROGRAM

I LANDING



EASY TO USE
HELP
POTENTIAL
DATABASE

II SIGN UP



III SIGNUP-VERIFICATION



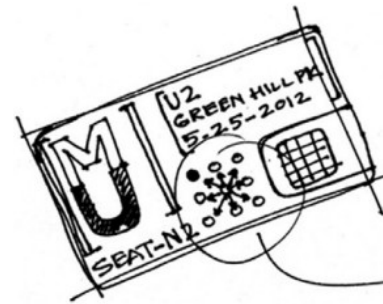
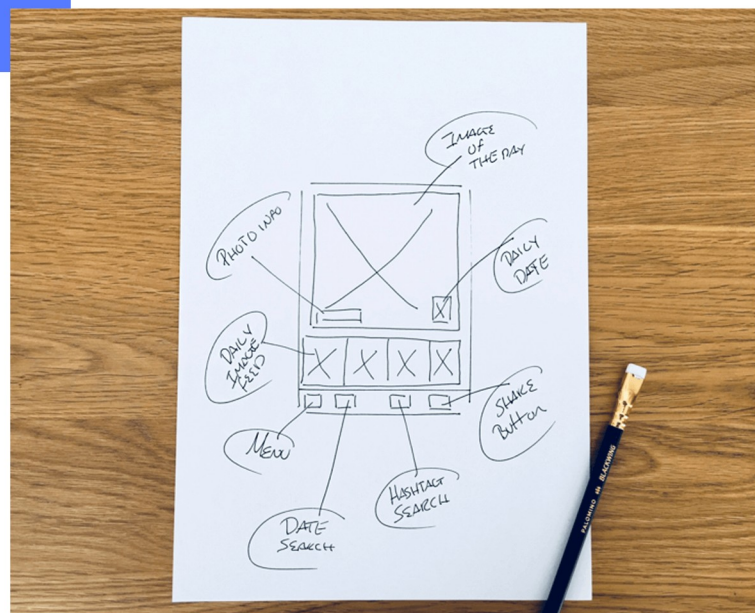
EMAIL

CONFIRM
VERIFIED/NO

V DASHBOARD



PAYOUT
FLOW



SEAT FINDING ASSISTANCE
SYSTEM.

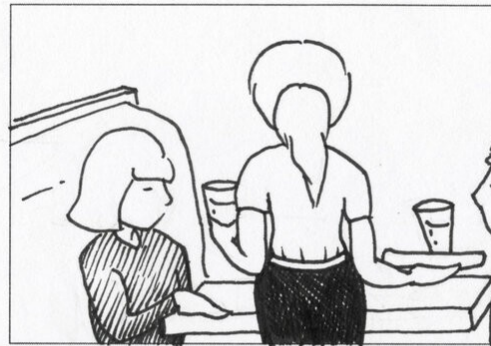


THE LIGHT BLINKS
TO INDICATE THE
DIRECTION OF YOUR SEAT.

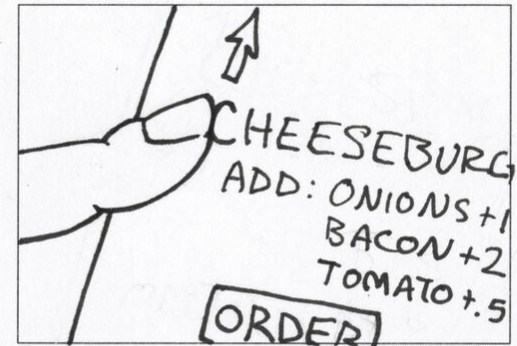
Ordering



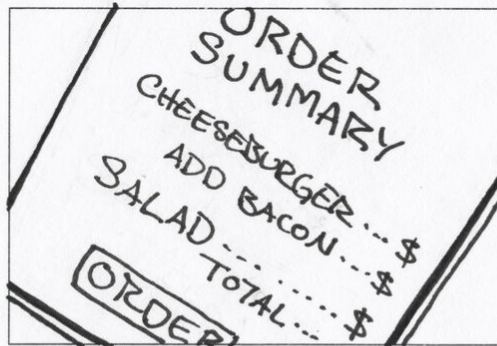
Server sees arrival of new table on screen in Service Station $\Rightarrow$ Brings water



"Hi guys. I'm Sheila. Just press HELP if you need anything."



"This looks good." $\Rightarrow$
Select food items $\Rightarrow$ Mod:fy
 $\Rightarrow$ place order



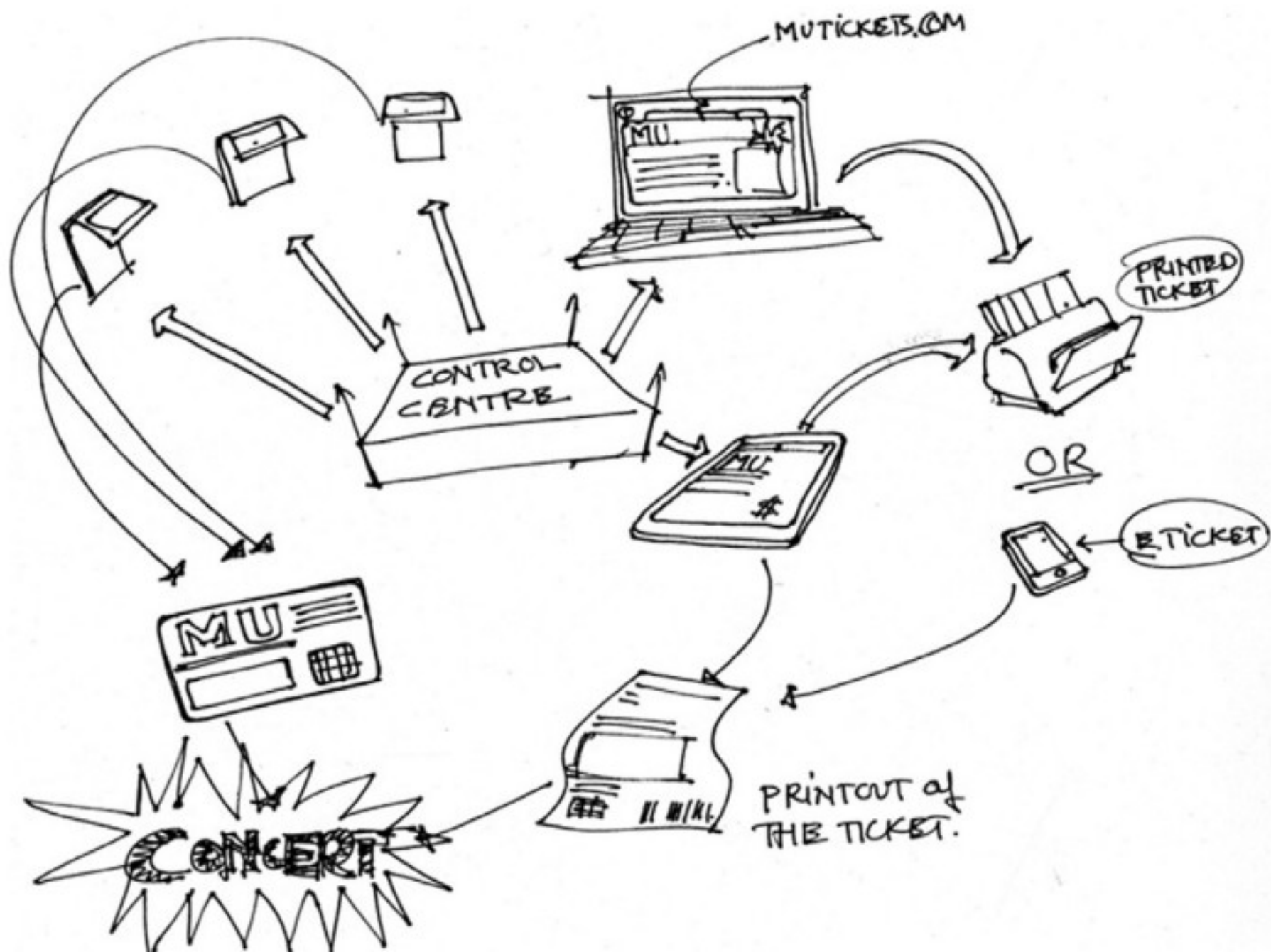
Review order $\Rightarrow$ Place Food order $\Rightarrow$ Sent to Kitchen

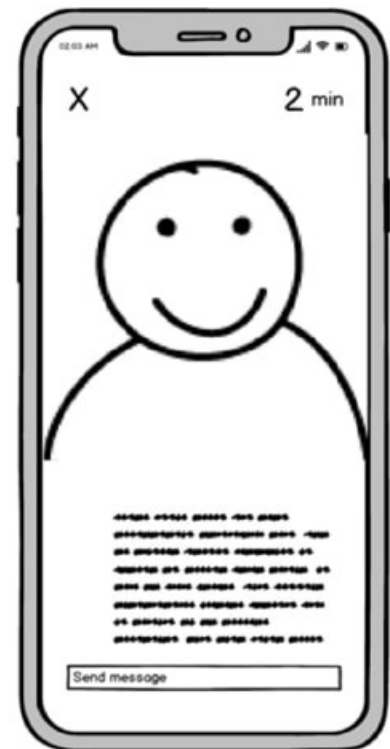
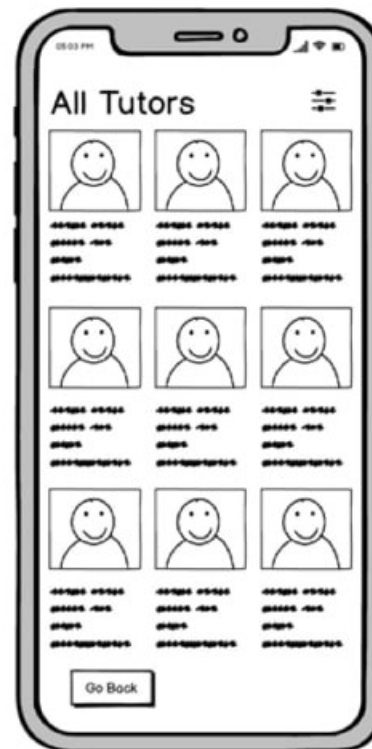
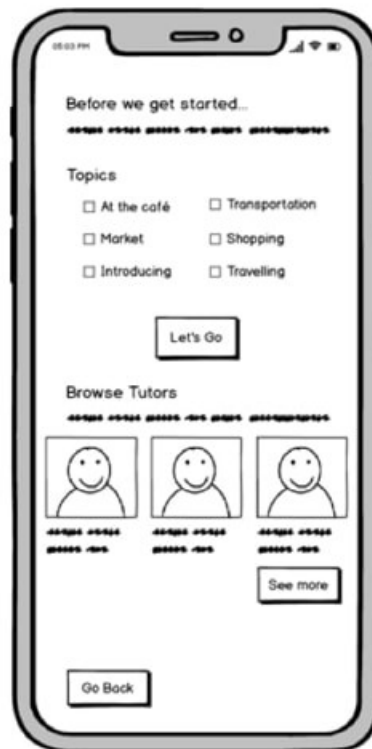
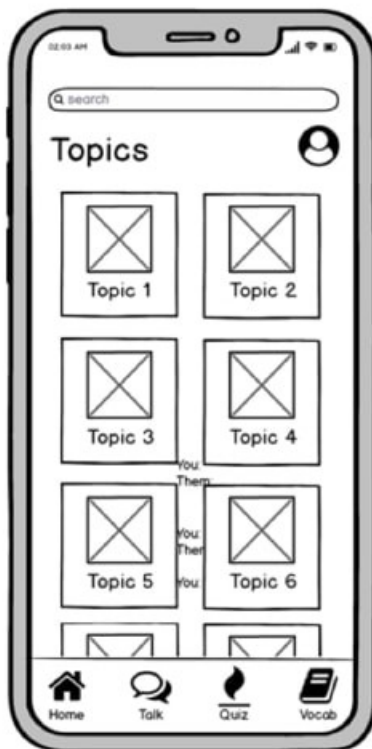


"Order in. Cheeseburger with bacon and a salad for table 13." Cook can track orders on monitor.



Server sees table 13's food is ready $\Rightarrow$ picks up plates from kitchen



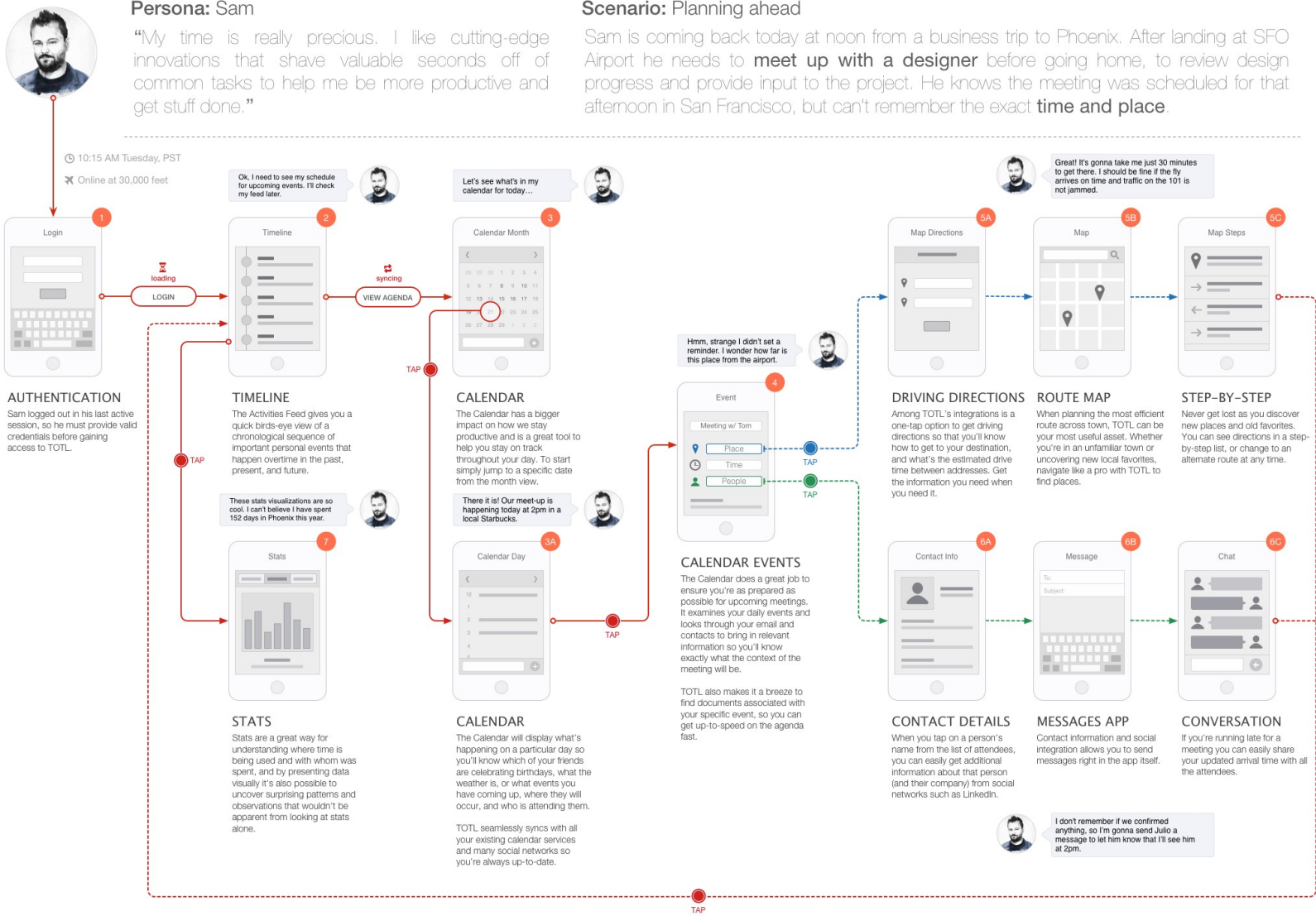


02

USER JOURNEYS

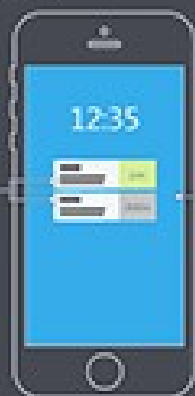
A user journey is a series of steps which represent a scenario in which a user might interact with the product we are designing.

User journeys are a great way to communicate and demonstrate the vision of the project, they help us understand user behavior, and identify functionality requirements at a high level to enable key user tasks.





• Start 0



• The First Screen 1.1



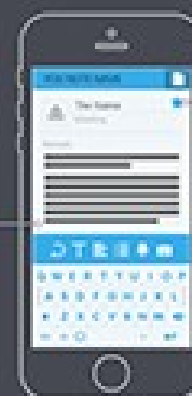
• Note Screen



• Remind Screen



• Remark

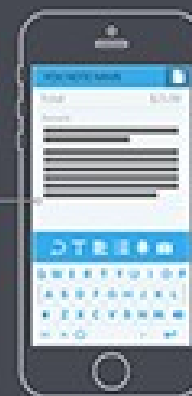
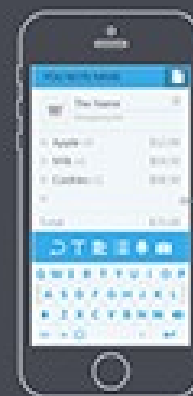
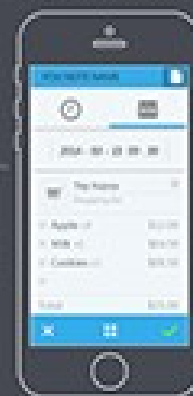


Important

• The Search Screen 1.5



• The List/Schedule Screen 2.1



Keyboard shortcuts: **←** and **→** to navigate, **ESC** to close, **+** and **-** to zoom, **A** to annotate.

Fixed Header



0.0x

Hotspot Destination
08-My Profile.psd

Screen Transition
None

Action
Click or tap



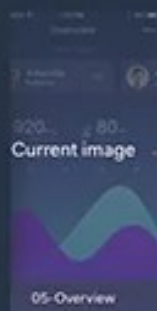
← Link to last visited

🔗 Link to URL or embed

📌 Scroll to anchor

📄 Keep scroll position

🖼️ Layer an image



05-Overview



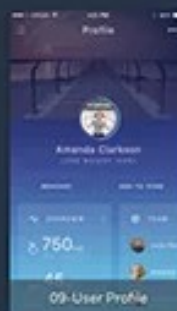
06-Overview (Change...)



07-Menu



08-My Profile



09-User Profile



10-Inbox



11-New Message



12-Notifications



Webcam

Facilitator's
Computer

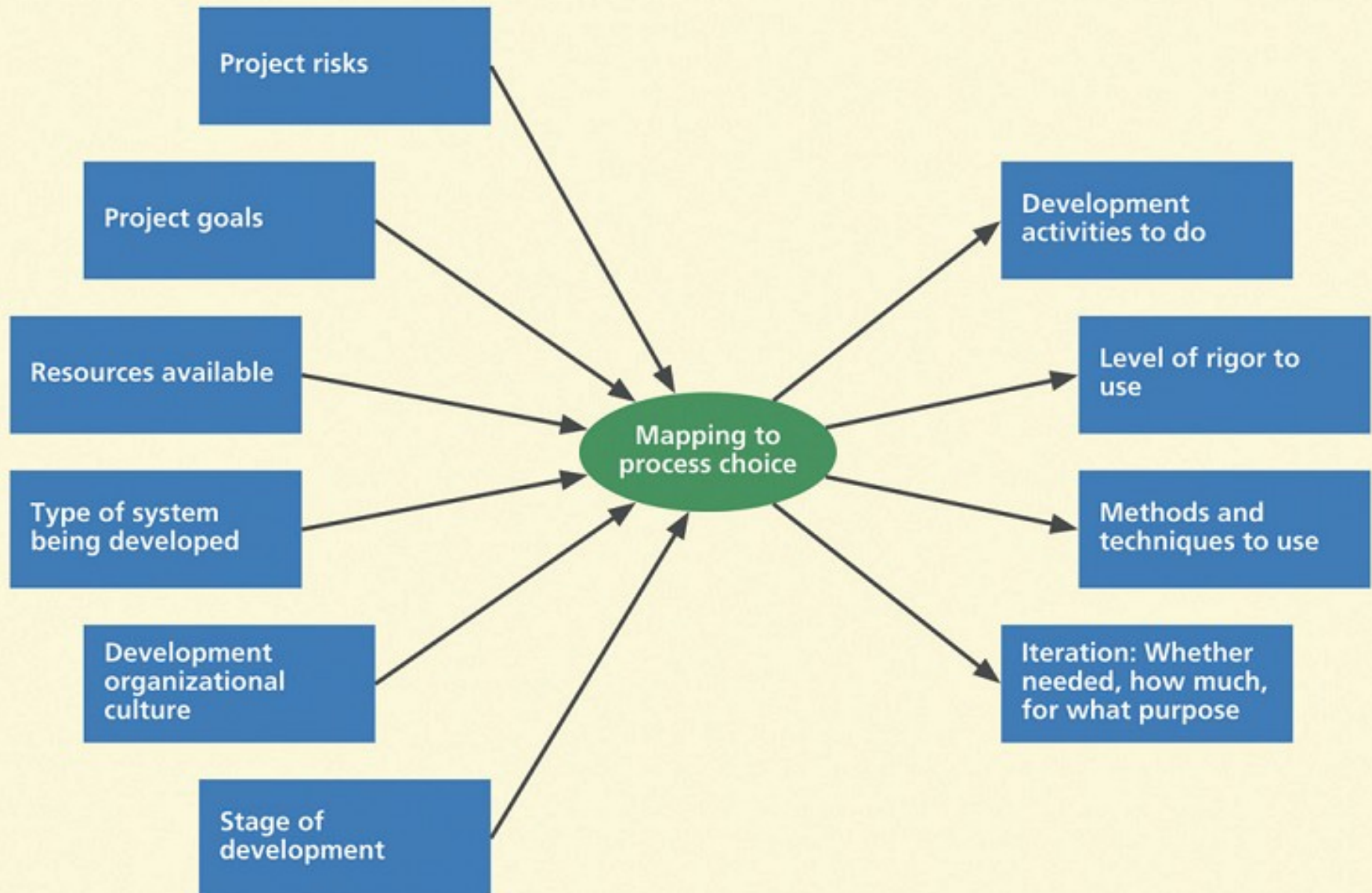
Mobile
Testing Sled

Mobile
Application

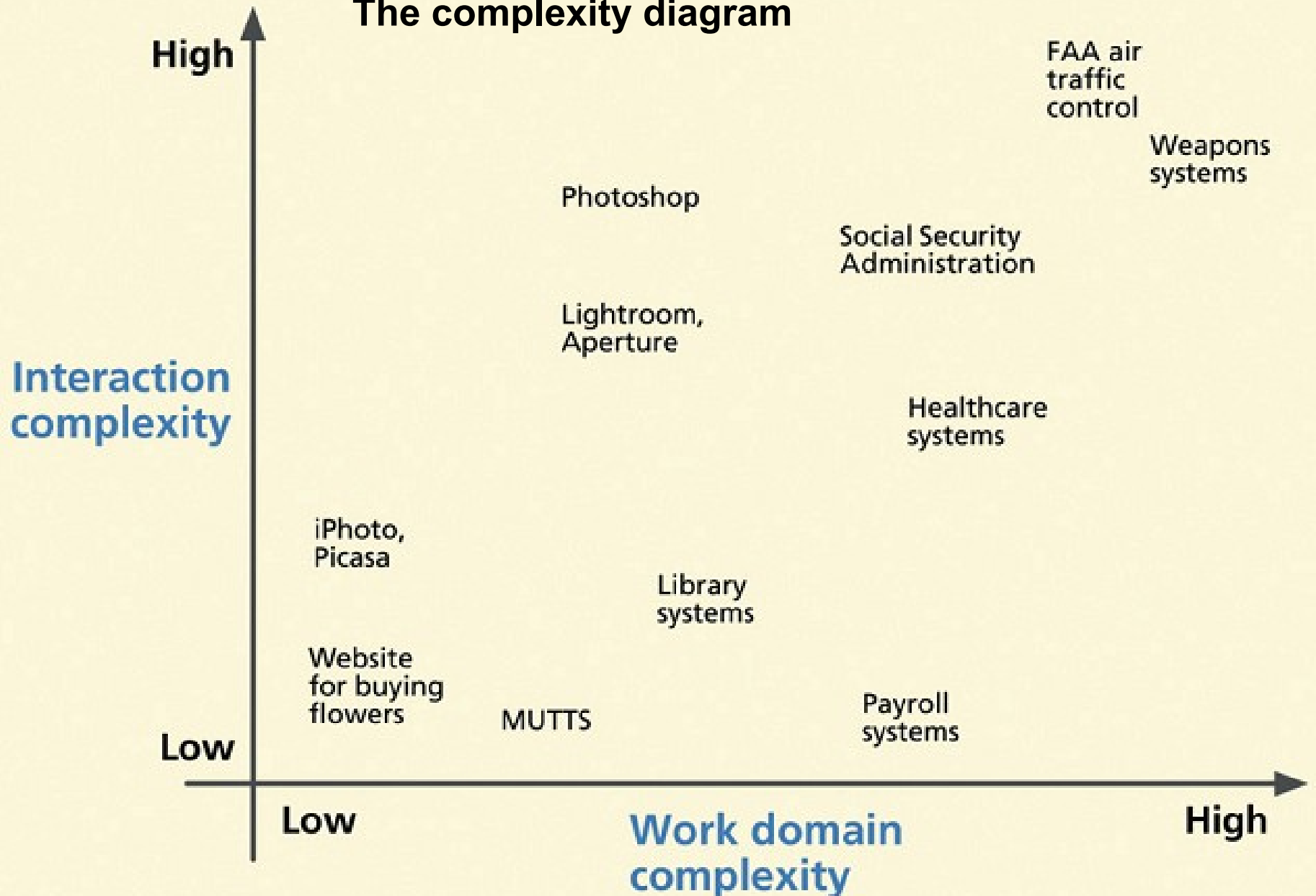
Test
Participant

Project Parameters

Process Parameters



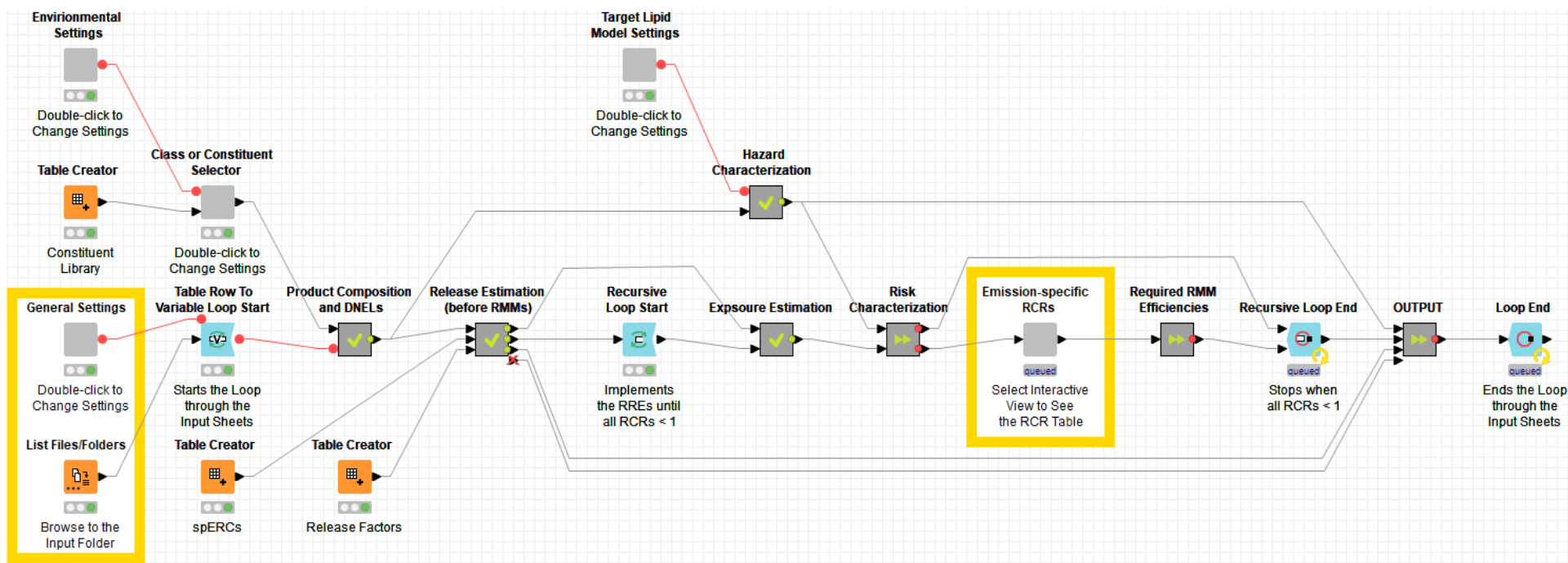
The complexity diagram

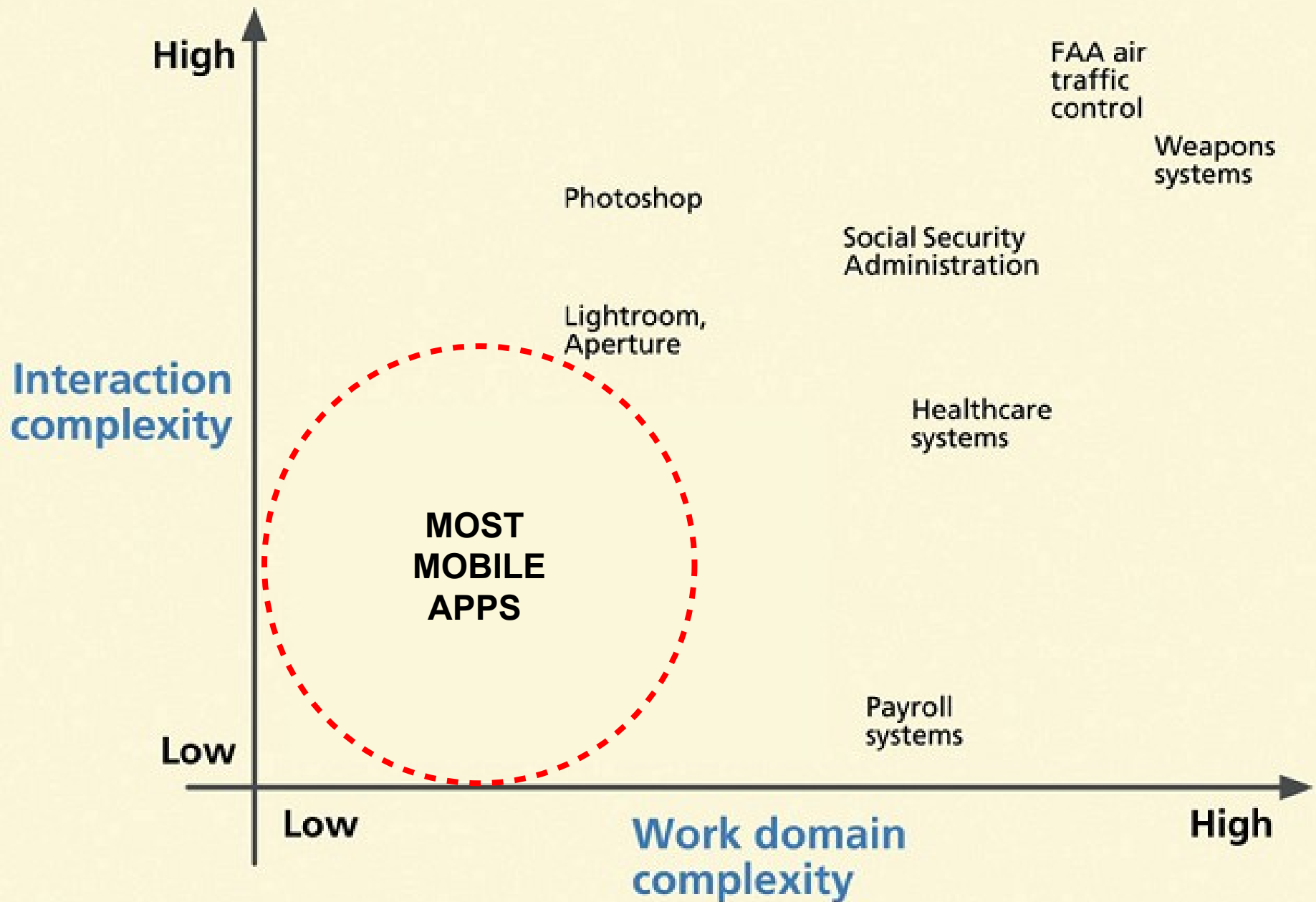


- Domain-complex software has complicated workflows that require a high degree of technical knowledge



- Interaction complexity is about the intricacy or elaborateness of user actions necessary to accomplish tasks with the software

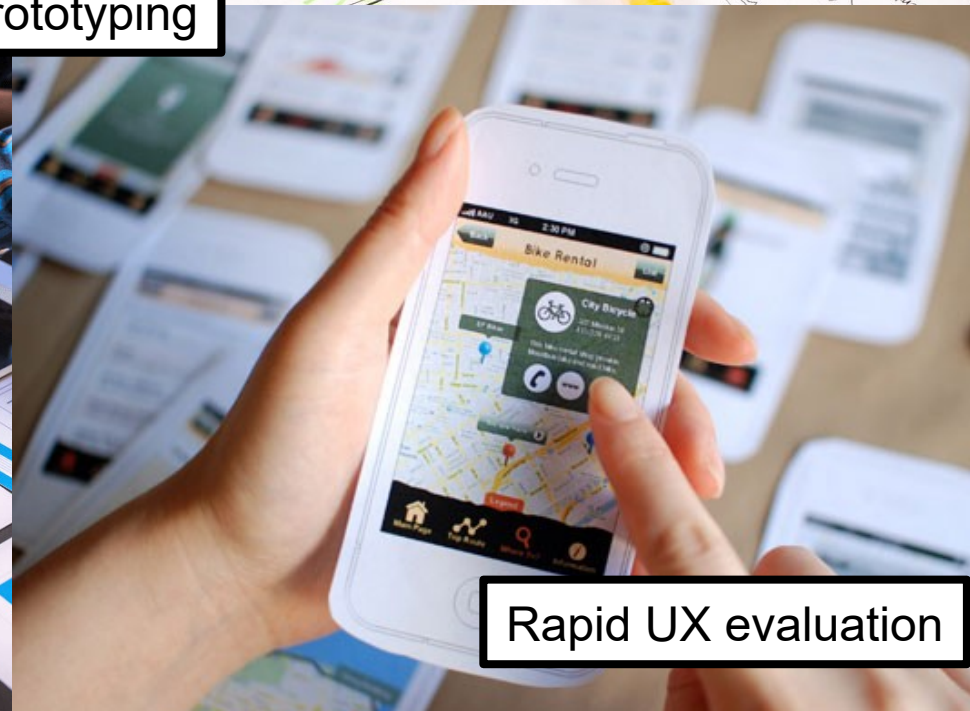




Agile design and development process



Rapid prototyping



Rapid UX evaluation

- The approach we follow to design mobile apps involves three activities/steps:
 - **Envision**: create the vision for the app
 - **Prototype**: create rapid app prototypes to explore possible designs
 - **Evaluate**: test the prototypes with potential users and iterate on the most promising design
- Each step can involve different techniques used more or less rigorously depending on the complexity of the project